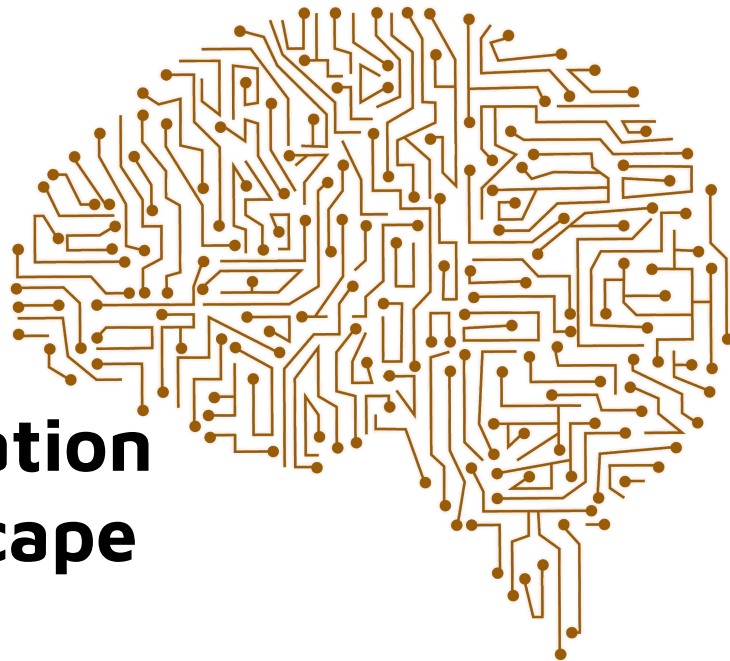




Dissertation Presentation

EEG Analysis for Authentication and Affect-Guided Soundscape Exploration

脳波解析による認証と立体音響の探索

Presented by Isuru Jayarathne (d8191101)

Supervised by Prof. Michael Cohen

Evaluators: Prof. Zhao Quangfu, Prof. Chen Wenxi, Prof. Yaguchi Yuchi

October, 2020



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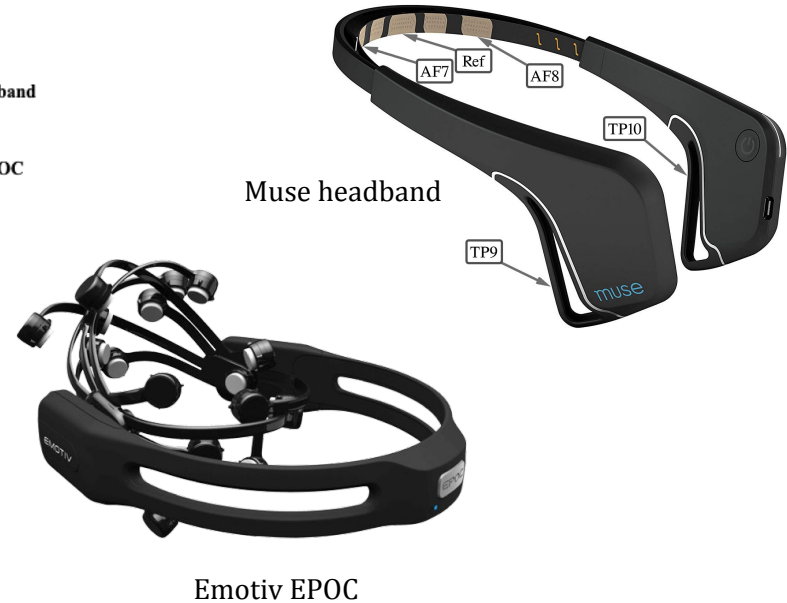
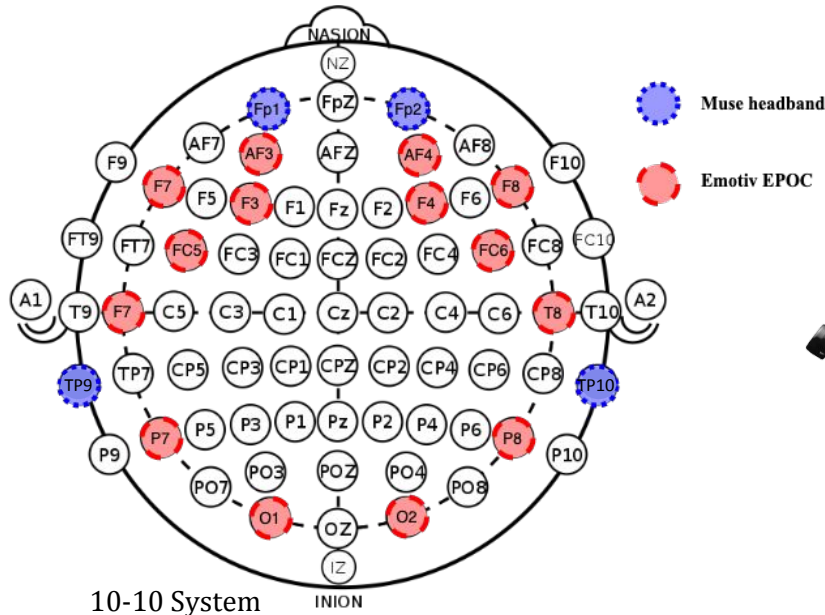
Introduction

- Main focus of both projects is analysis of EEG for idiosyncrasy using machine learning approaches.
- Investigation of subjective features from EEG signals for biometric authentication
- Exploitation of EEG and reinforcement learning to enhance musical relaxation



Introduction

- Electroencephalography (EEG)
 - Electrophysiological monitoring method to record electrical activity of one's brain
 - Usually, 10-20, 10-10, or 10-5 electrode placement systems are used.





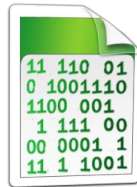
EEG-Based biometric identification and authentication





Introduction

- Authentication: process to verify that user has privilege to access secure information
- Identification: process of determines identity, simply 1-of-n





Introduction

- Authentication strategies
 - Knowledge: “Something one knows” (password, PIN, ...)
 - Possession: “Something one has” (ATM card, passport, ...)
 - Biometric: “Something one is”
 - Physiological (fingerprint, voice print, iris scan, ...)
 - Behavioral (gait, keystroke dynamics, ...)





EEG as biometric

- A predefined brain state which authenticates a user is impossible to evoke by insistence or coercion.
- No non-living brain can produce an EEG, which prevents breaching an authentication system by assassinating a user.
- Even though many other biometric traits are visible, EEG biometrics are invisible, making them uncapturable by other parties.



Literature Review

Authors	Channels	Subjects	Task	Extracted features	Classifier	Avg. accuracy
Ruiz-Blondet et al.	3	50	Visual stimulation of 400 images	Event-Related Potentials (ERP)	Normalized cross-correlation	100%
La Rocca et al.	64	108	Relaxation with opened eyes and closed eyes	Power Spectral Density (PSD), Spectral Coherence (COH)	Mahalanobis distance-based classifier and match-score fusion	100%
Chen et al.	16	29	Rapid Serial Visual Representation (RSVP)	Point-biserial correlation coefficients, Fisher's transformation	Linear Discriminant Analysis (LDA)	100%
Chuang et al.	1	15	7 tasks	Cosine similarity of the vector representation	k-Nearest Neighbors (k-NN)	99%
Palaniappan	6	6	5 tasks	AR, SP, IHDP, IHLC	LDA	100%
Ashby et al.	14	5	4 tasks	AR, SP, IHPD, IHLC, PSD	Support Vector Machine (SVM)	100%
Riera et al.	4	-	Relaxation	AR, FFT (EEG+ECG)	Fisher's Discriminant Analysis (FDA)	98%
Yeom et al.	18	10	Images of faces including self-face as visual stimulation	Difference of average signals	SVM	86%



Literature Review

- Most approaches are not user-friendly, even though perfect accuracy has been achieved.
- Problems with current systems
 - Require many trials to train ✓
 - Whole system must be retrained when adding a new user ✓
 - Task complexity ✓
 - Instability of brain signal pattern ✗
 - Use of clinical-grade devices (expensive) ✓

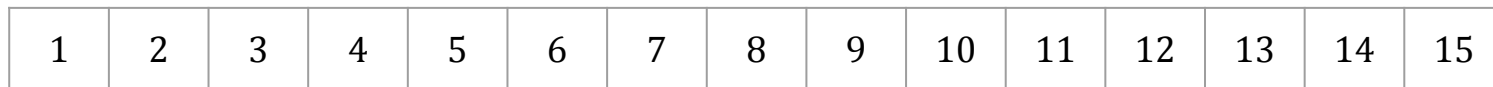




Data collection

- Dataset
 - 12 subjects
 - 15 trials conducted for each subject
 - 3 phases for each subject (540 trials totally)
 - Random 4-digit numbers used for visual stimulation
 - Data captured with 128 Hz sample rate

Single Subject



Each Trial



Relaxation

Visual
Stimulation

Mental Recall

10 s

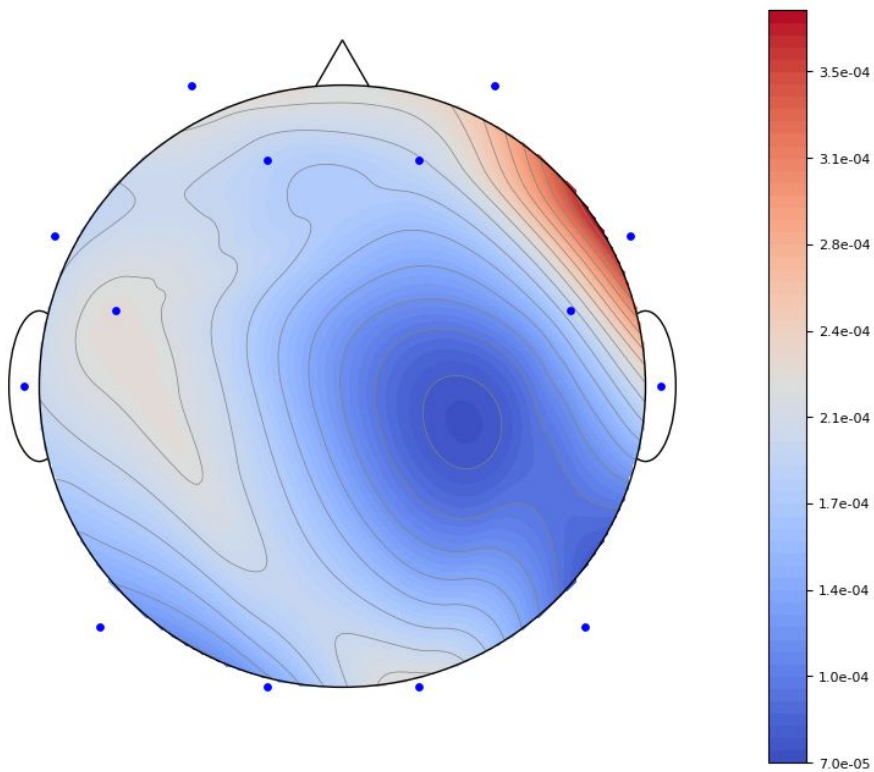
5 s

10 s



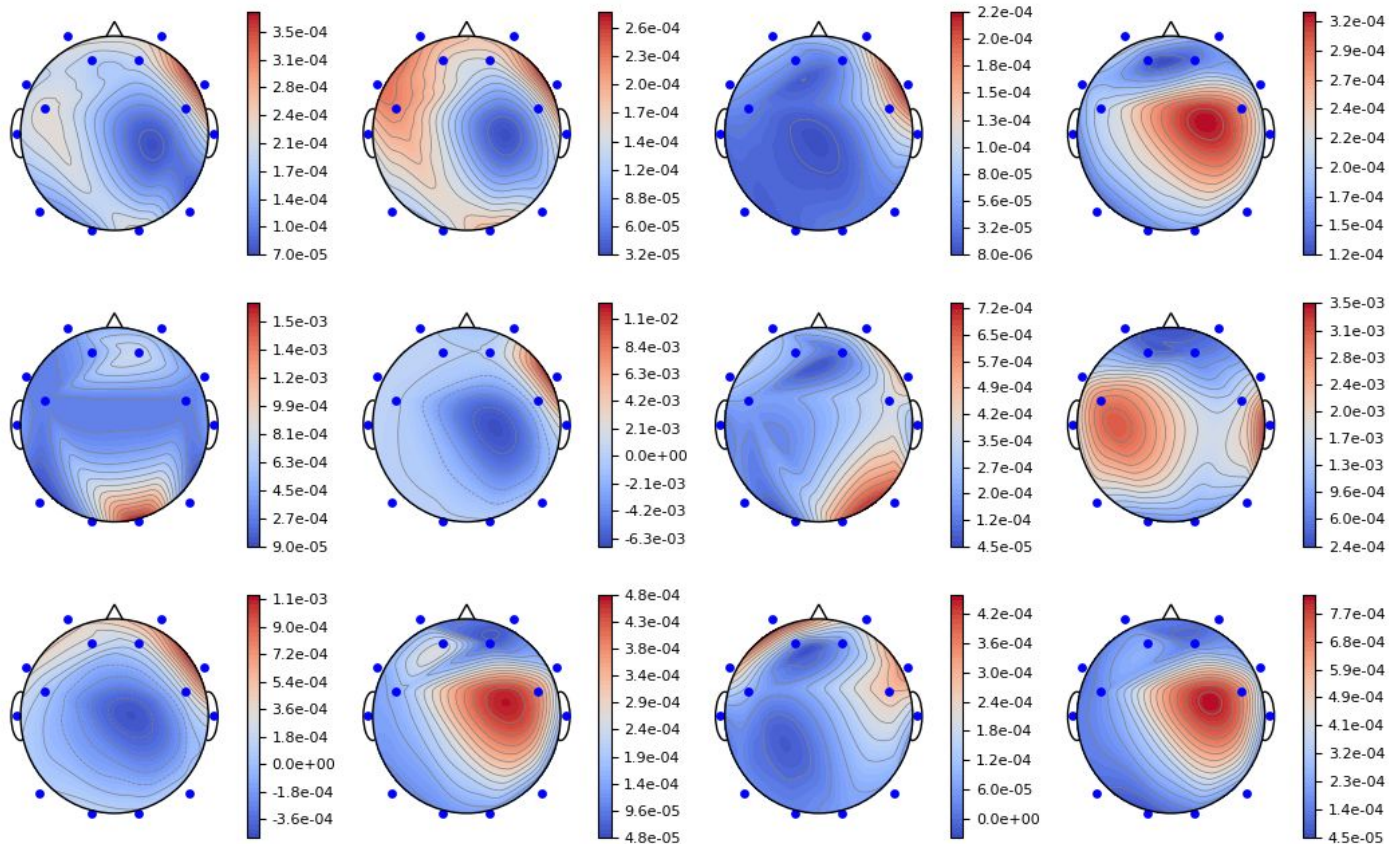
Data visualization

- Topographic plot of power distribution



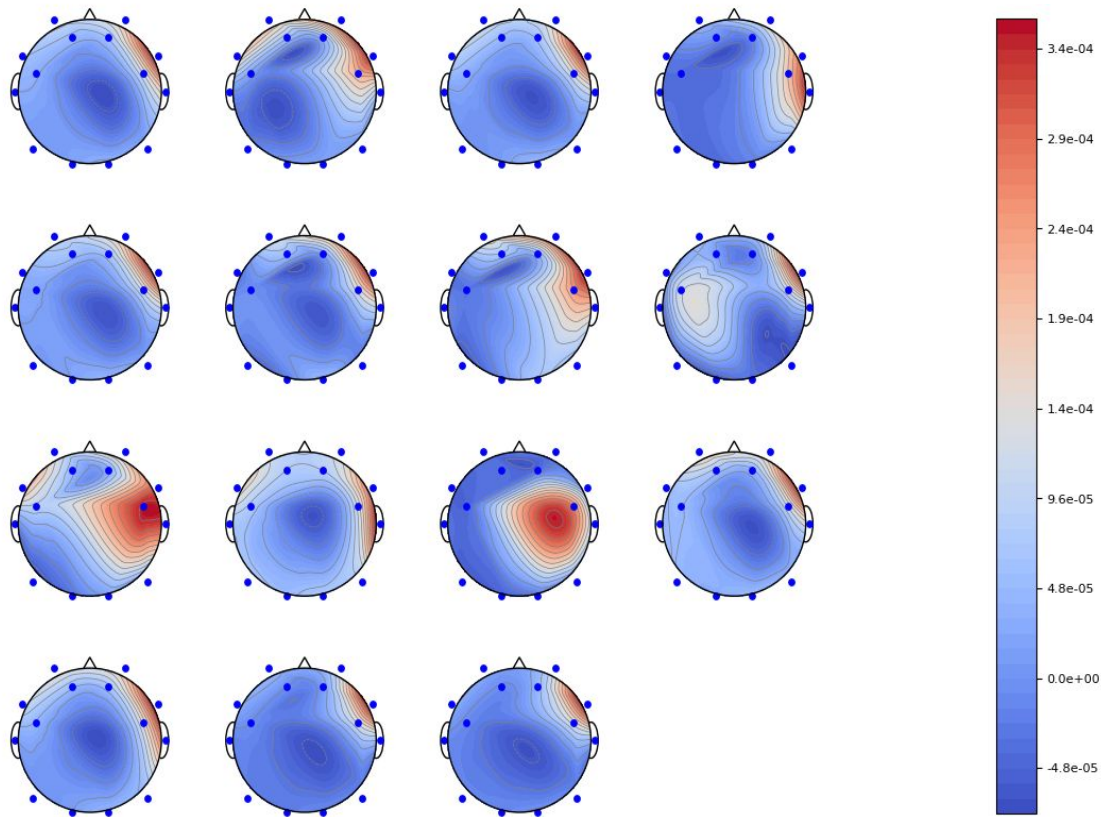
Data visualization

- Topographic plots for 12 subjects (Relaxation phase)



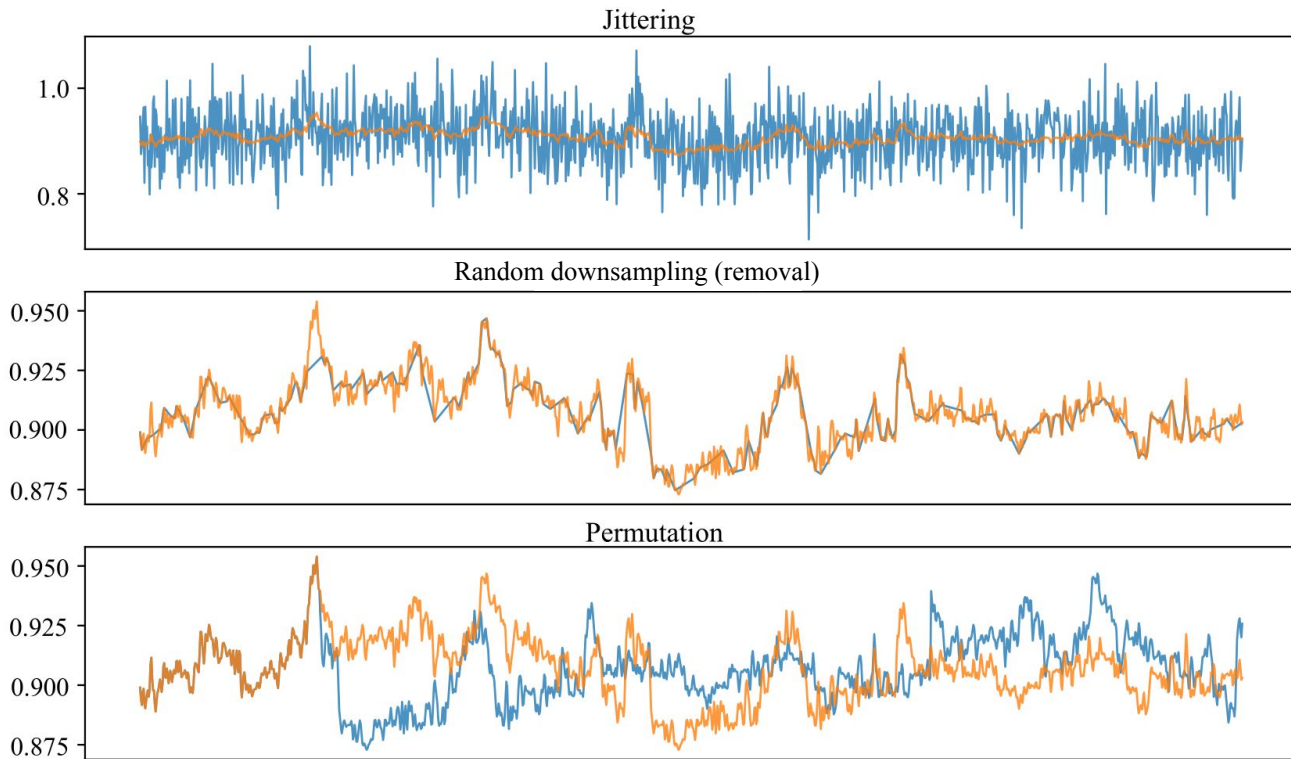
Data visualization

- Topographic plots of trials for Subject 1 (Delta frequency band (0.5–4 Hz) during Relaxation)



Feature extraction with IHAR: Data augmentation

- Visualization of raw and augmented signals subject to jittering, random sampling, and permutation techniques.

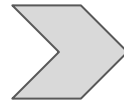




Feature extraction with IHAR

- Derived a new feature, dubbed Inter-Hemispheric Amplitude Ratio (IHAR)
- At least 2 electrodes are needed
- Derivation process of the feature:

Moving Average (with
window size 32)

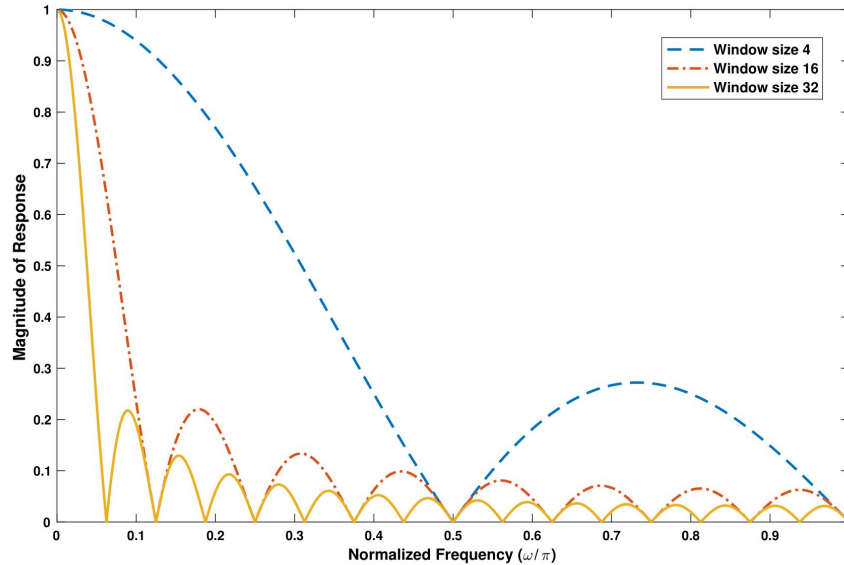


Taking ratio of
corresponding left and right
channels

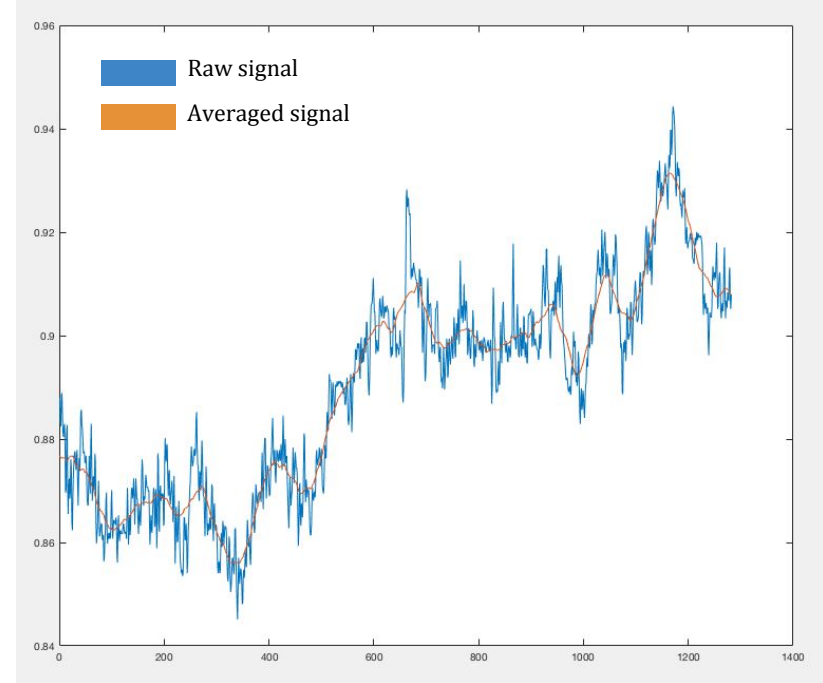


Feature extraction with IHAR

- Moving average (MA) filtering



Frequency response of MA filter

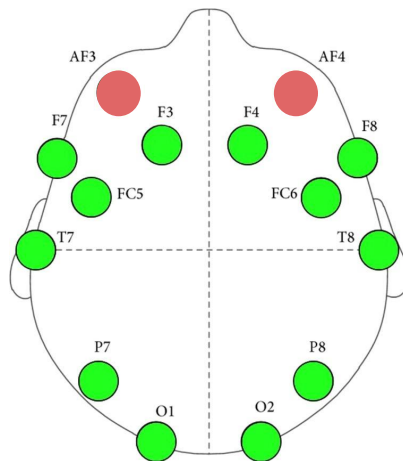


After taking moving average



Feature extraction: IHAR

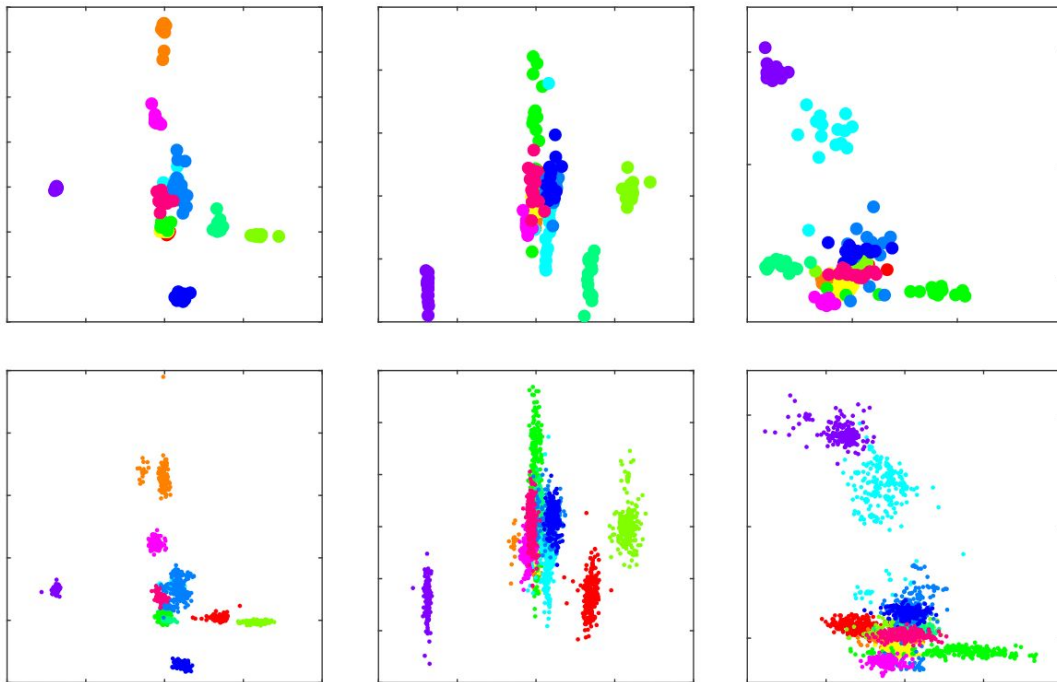
- Since electrode placement is symmetric across median or sagittal plane, IHAR feature can be calculated.
- Ratio is calculated with corresponding electrodes of opposite sides' brain lobes.



$$x_{\text{IHAR}} = \frac{x_{\text{AF3}}}{x_{\text{AF4}}}$$

Feature extraction with IHAR: Data augmentation

- Distribution of features before and after augmentation, using color to distinguish respective subjects





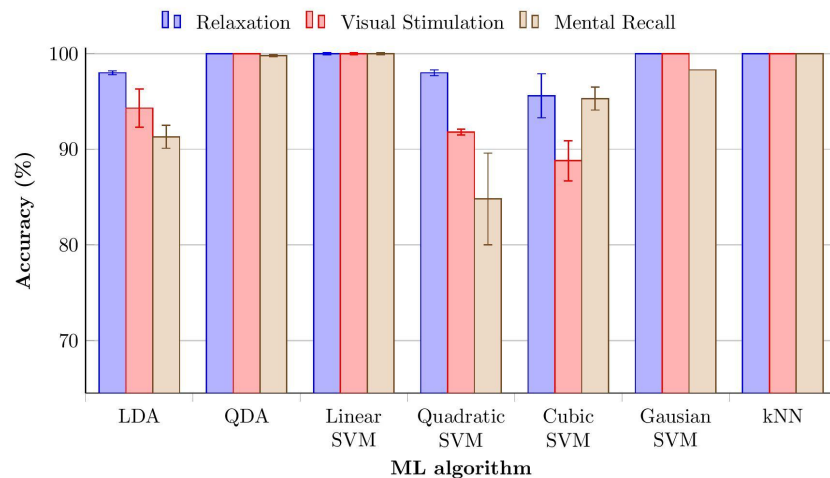
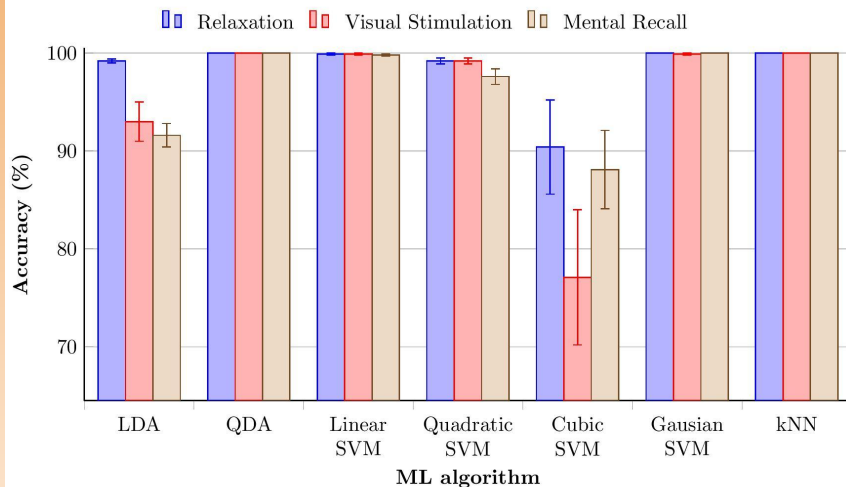
Feature extraction with IHAR: Evaluation

- Performance evaluation of the derived feature
 - Derived feature was tested with several machine learning algorithms.
 - 10 original trials were used as training set with 10 generated instances of each augmentation technique $((10 \times 10 \times 3 + 10) \times 12 = 3720$ in total).
 - The remaining 5 original trials from each subject were used as testing set (60 in total).
 - Further, machine learning algorithms which showed highest accuracies were tested by reducing the number of electrodes.



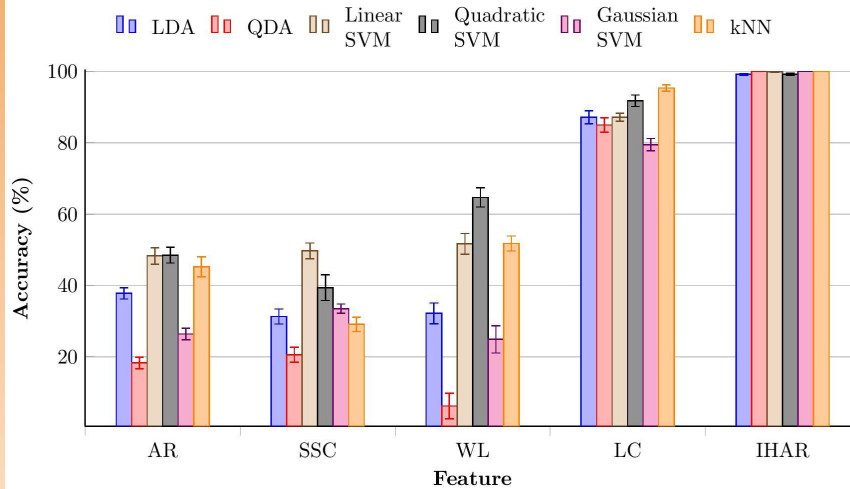
Feature extraction with IHAR: Evaluation

- Accuracies for all electrodes with three different phases

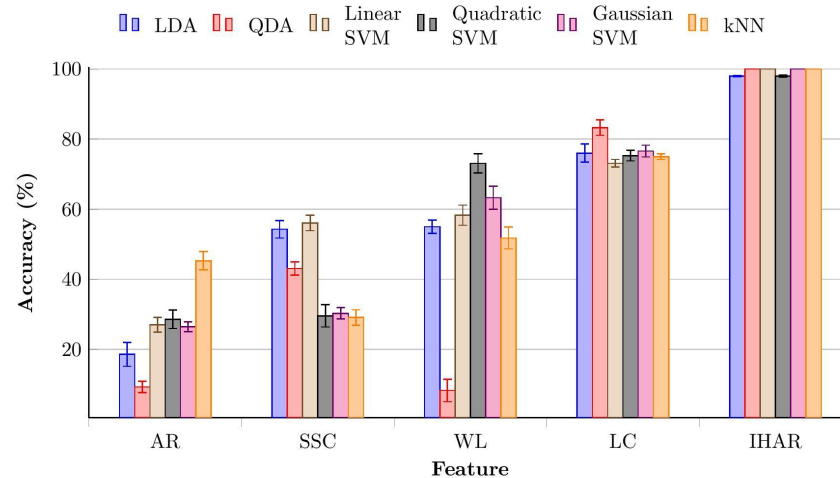


Feature extraction with IHAR

- Comparison of accuracy across features



Validation accuracies

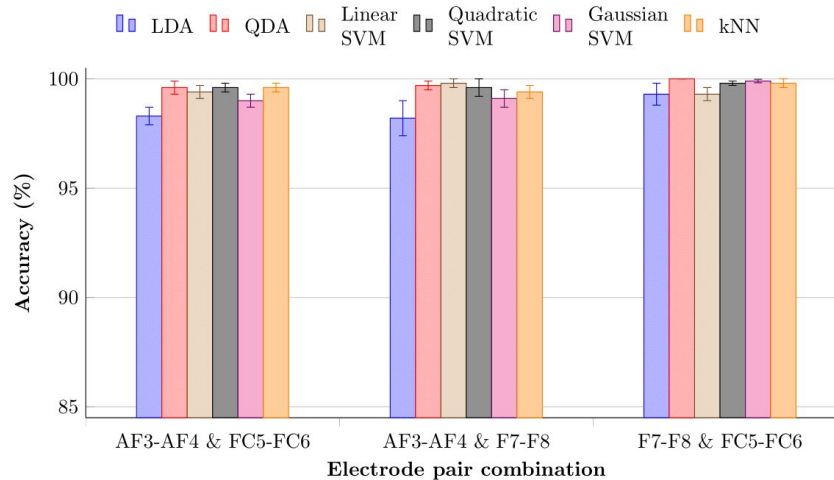


Testing accuracies

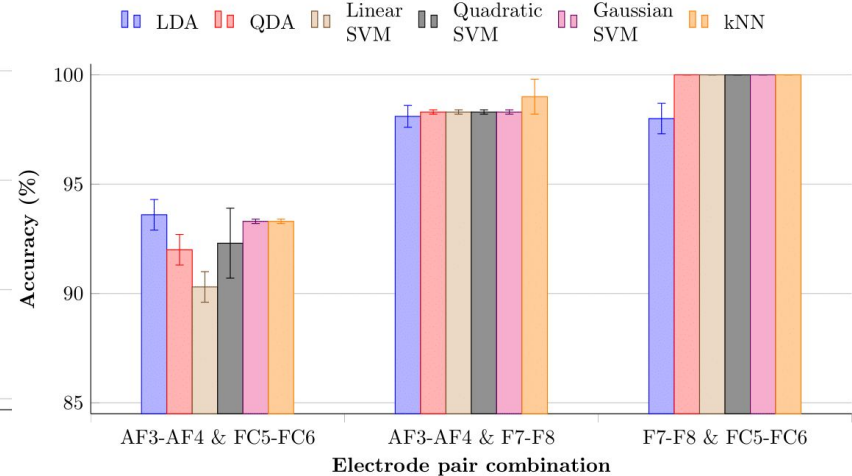
AR: Autoregressive coefficients, SSC: Slope sign change, LC: Laterality coefficient, WL: Waveform length

Feature extraction with IHAR

- Behavior of accuracy while reducing number of electrodes for relaxation



Validation accuracies

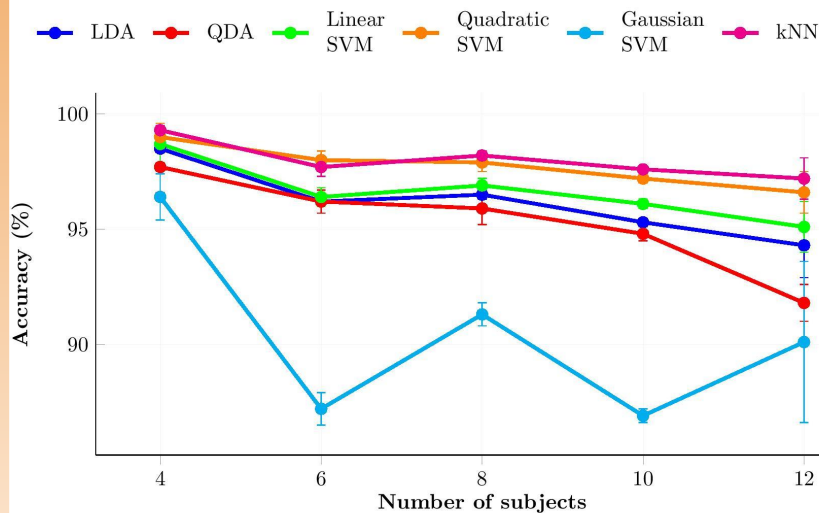


Testing accuracies

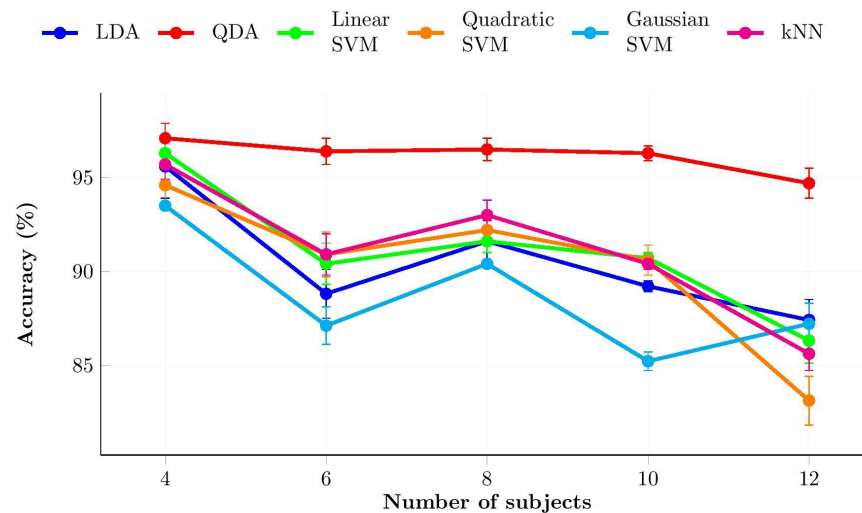


Feature extraction with IHAR

- Variation of accuracy when varying number of subjects



Validation accuracies



Testing accuracies





Feature extraction with IHAR

Conclusion

- High accuracies can be achieved (average AUC = $0.98 \sim 1$),
 - Using as few as 4 electrodes
 - Signals from frontal cortex (F7 & F8, FC5 & FC6)
- Relaxation is the best phase among three phases regarding both user friendliness and accuracy.
- According to the results IHAR can be used to discriminate people using captured EEG signals.





Autoencoder-based one-class classification

Motivation

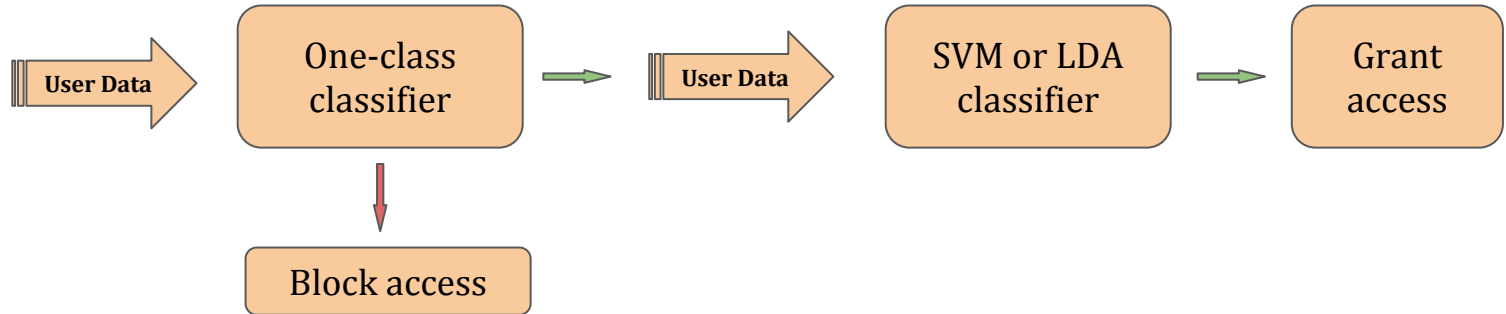
- Problem of using traditional machine learning algorithms (LDA, SVM, kNN, etc.) in authentication context
 - Cannot filter imposters
 - System must be retrained when each new user is added





Autoencoder-based one-class classification

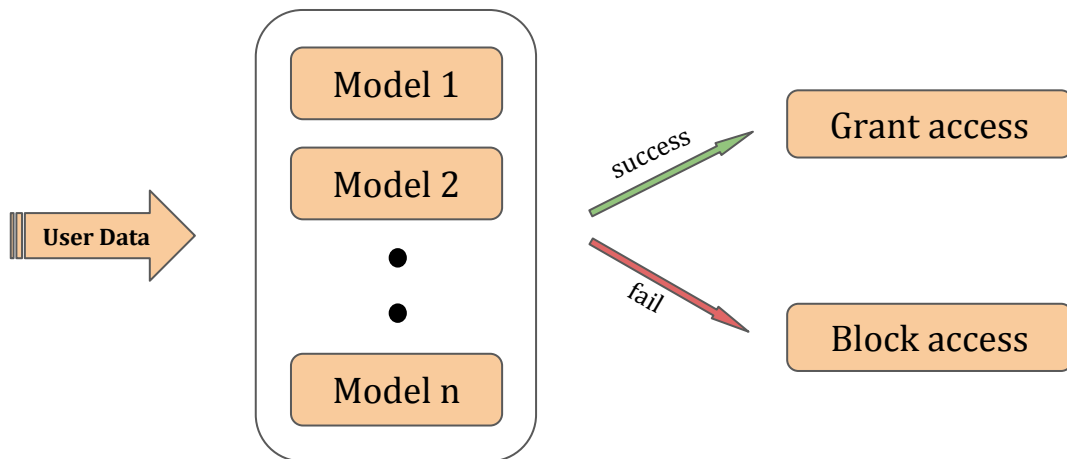
- Method 1: Treat all system users as single class and train one model. Then identify confirmed user by LDA or SVM





Autoencoder-based one-class classification

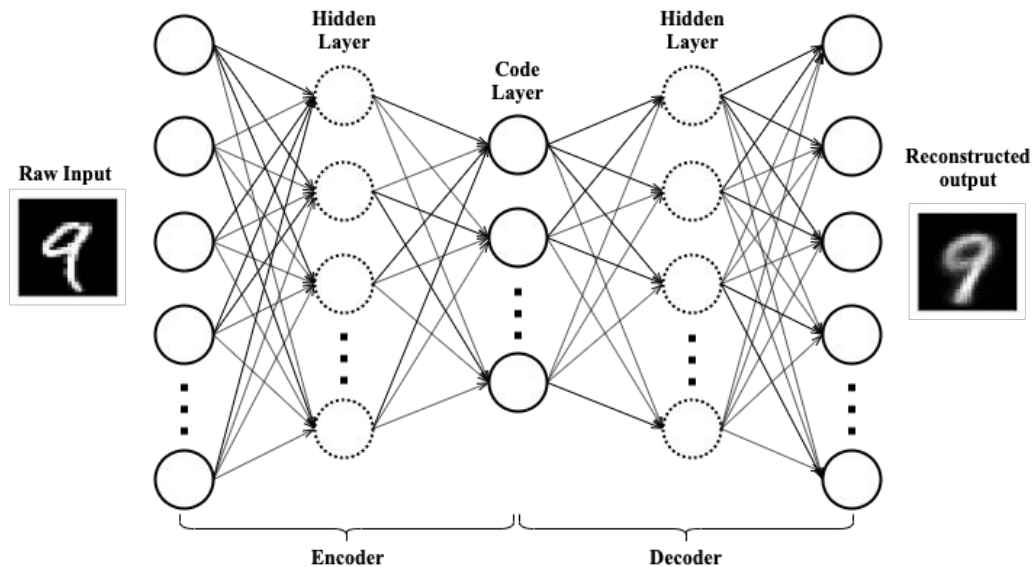
- Method 2: Train a separate model for each system user, and check across all trained models when authenticating.



Autoencoder-based one-class classification

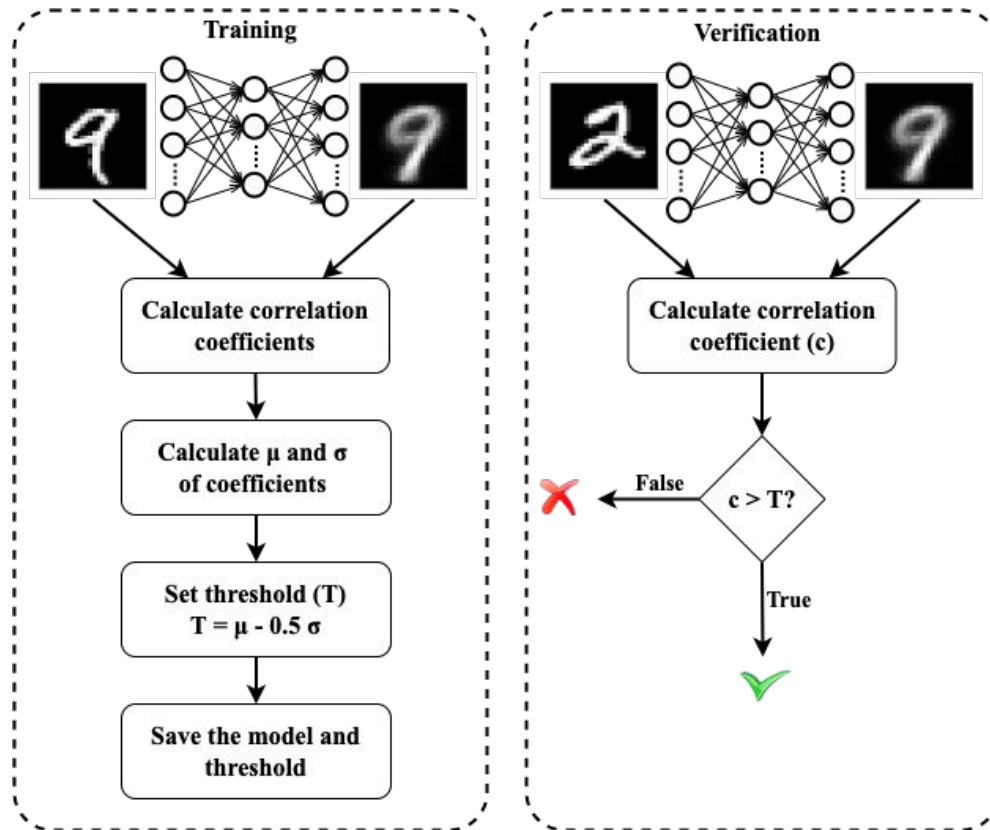
Introduction

- Autoencoder is a dimensionality-reduction algorithm.
- Compress (encode) and reconstruct (decode) data
- ANN with fully-connected layers



Autoencoder-based one-class classification

Flow of training and verification phase





Autoencoder-based one-class classification

Pilot experiment

- Test with MNIST (handwritten characters) dataset
 - 60,000 training images
 - 10,000 testing images
- Each image has 784 (28×28) pixels.



Sample images of 10 digits



10 samples of digit 9





Autoencoder-based one-class classification

Pilot experiment

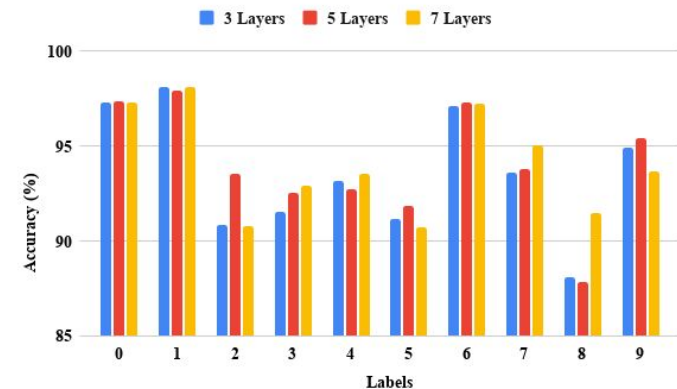
- 6 different autoencoder configurations were tested.
- Input and output layers have equal numbers of nodes (784).
- Number of hidden layers and number of nodes in code layer are varied:
 - 3 layers, 64 nodes
 - 5 layers, 64 nodes
 - 7 layers, 64 nodes
 - 3 layers, 128 nodes
 - 5 layers, 128 nodes
 - 7 layers, 128 nodes



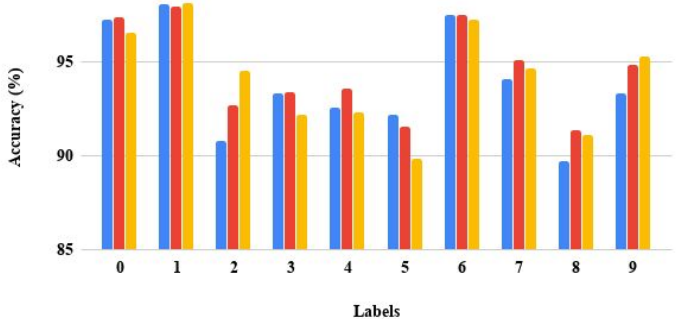
Autoencoder-based one-class classification

Pilot experiment results

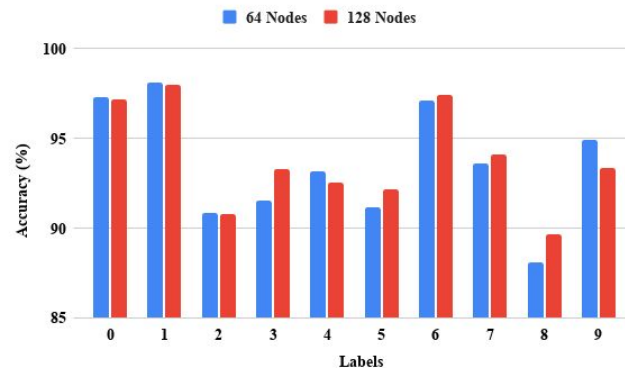
- No substantial difference in accuracies



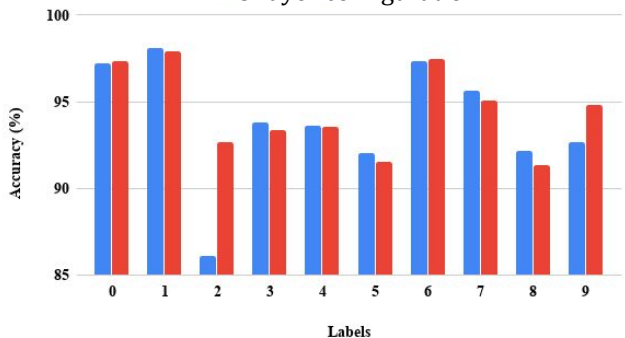
64-nodes configuration



128-nodes configuration



3-layer configuration



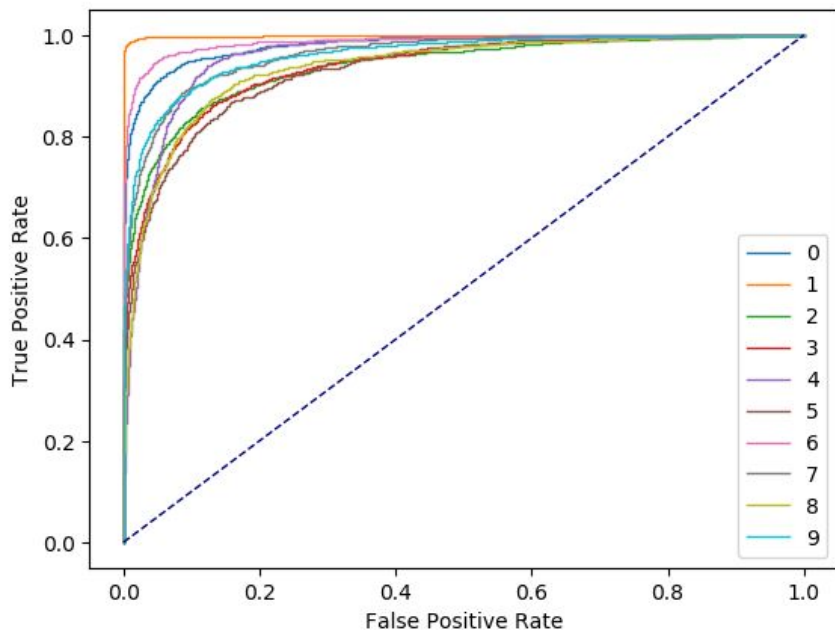
7-layer configuration



Autoencoder-based one-class classification

Pilot experiment results

- ROC curve of calculated correlation coefficient values
- No considerable difference in means of AUC values



ROC curve of 3 Layers, 128 nodes

Configuration	Mean AUC value
3 Layers, 64 nodes	0.9594
5 Layers, 64 nodes	0.9581
7 Layers, 64 nodes	0.9575
3 Layers, 128 nodes	0.9622
5 Layers, 128 nodes	0.9586
7 Layers, 128 nodes	0.9617

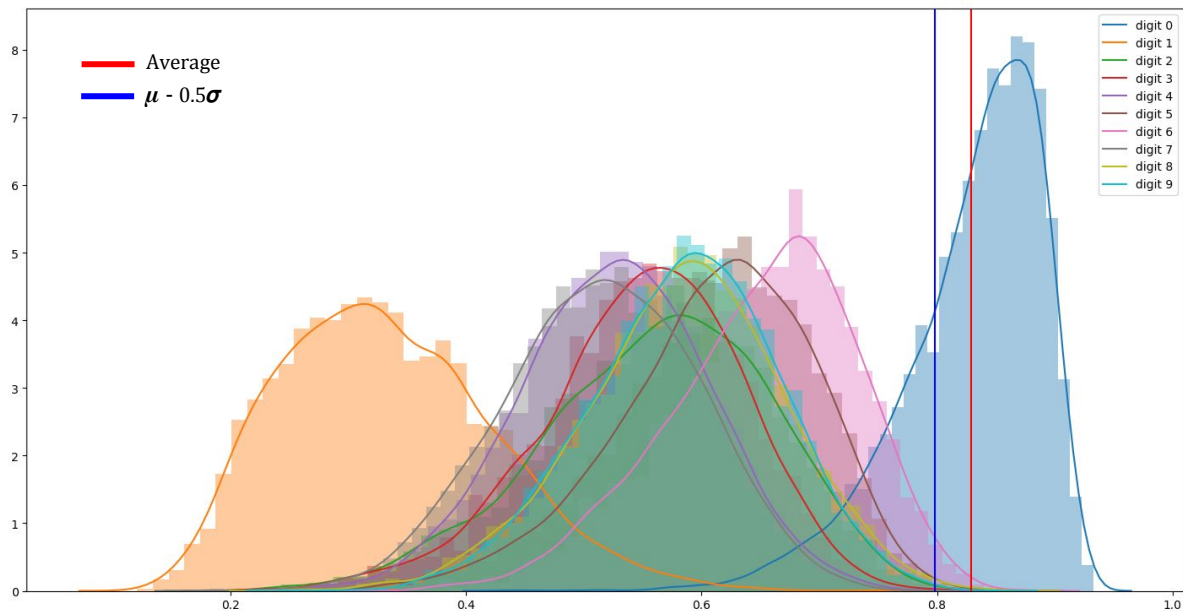
Comparison of AUC values



Autoencoder-based one-class classification

Pilot experiment

- Distributions of correlation coefficients of each digit after training the model for digit 1





Autoencoder-based one-class classification

Experiment with EEG dataset

- 3 layers, code layer of 128 nodes configuration was used.
- Same EEG dataset was used with 10 training trials and 5 testing trials.
- Results of relaxation phase of all channels:

Metric	Raw signal	IHAR	PSD
Accuracy	95%	88%	91%
FPR	0%	5.76%	1.21%
FNR	60%	85%	95%
AUC	0.7	0.55	0.52



Autoencoder-based one-class classification

Experiment with EEG dataset

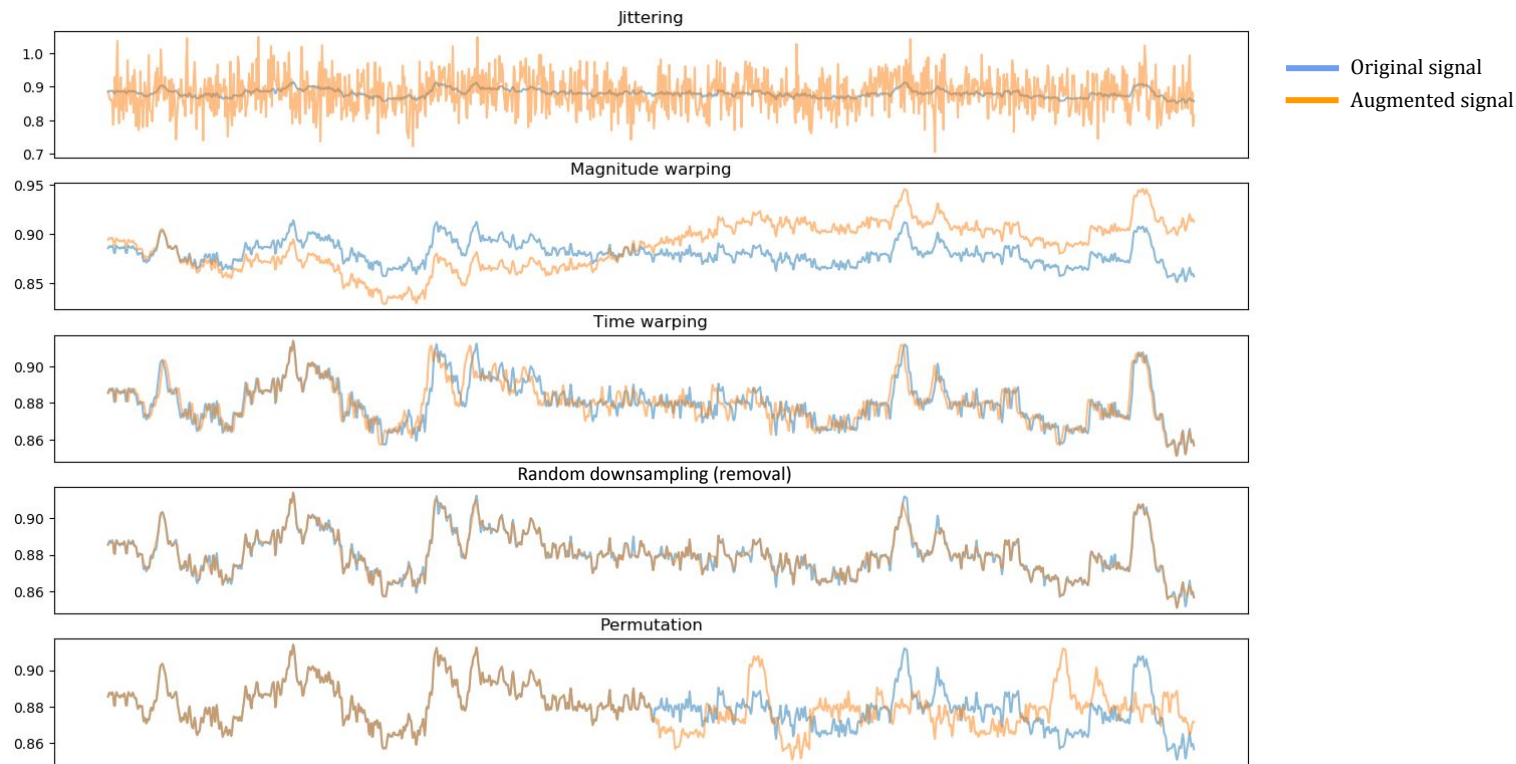
- To extend the training set, data augmentation was used.
- Following techniques were used:
 - Jittering
 - Magnitude warping
 - Time warping
 - Random downsampling (removal)
 - Permutation
- 10 samples were generated from each technique (510 training trials).





Autoencoder-based one-class classification

Experiment with EEG dataset





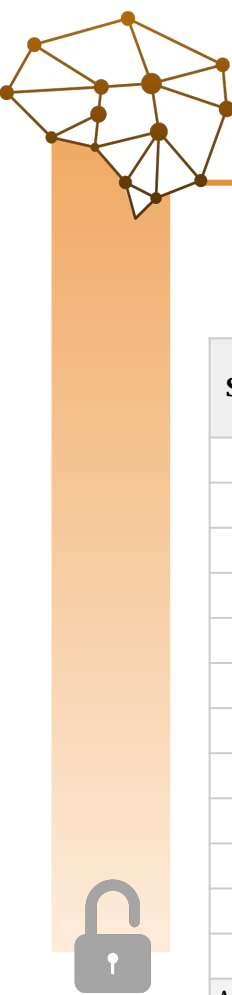
Autoencoder-based one-class classification

Experiment with EEG dataset

- Performance for raw signal dataset was significantly increased.
- IHAR and PSD showed lower performance than previous case (10 training trials).

Metric	Raw signal	IHAR	PSD
Accuracy	↓ 93%	85%	88%
FPR	↑ 7.27%	8.18%	5.91%
FNR	↓ 0%	85%	80%
AUC	↑ 0.96	0.53	0.57





Autoencoder-based one-class classification

Experiment with EEG dataset

- Subject-wise performance evaluation

Subject	Accuracy (%)			FPR (%)			FNR (%)			AUC		
	All	8	4	All	8	4	All	8	4	All	8	4
1	91.7	96.7	100.0	9.1	0.0	0.0	0.0	40.0	0.0	0.95	0.80	1.00
2	83.3	100.0	91.7	18.2	0.0	3.6	0.0	0.0	60.0	0.91	1.00	0.68
3	88.3	100.0	91.7	12.7	0.0	9.1	0.0	0.0	0.0	0.94	1.00	0.95
4	100.0	98.3	100.0	0.0	0.0	0.0	0.0	20.0	0.0	1.00	0.90	1.00
5	100.0	96.7	48.3	0.0	0.0	49.1	0.0	40.0	80.0	1.00	0.80	0.35
6	75.0	80.0	61.7	27.3	20.0	40.0	0.0	20.0	20.0	0.86	0.80	0.70
7	100.0	96.7	86.7	0.0	0.0	10.9	0.0	40.0	40.0	1.00	0.80	0.75
8	100.0	96.7	63.3	0.0	0.0	34.5	0.0	40.0	60.0	1.00	0.80	0.53
9	100.0	100.0	100.0	0.0	0.0	0.0	0.0	0.0	0.0	1.00	1.00	1.00
10	100.0	100.0	100.0	0.0	0.0	0.0	0.0	0.0	0.0	1.00	1.00	1.00
11	91.7	100.0	90.0	9.1	0.0	9.1	0.0	0.0	20.0	0.95	1.00	0.85
12	90.0	100.0	100.0	10.9	0.0	0.0	0.0	0.0	0.0	0.95	1.00	1.00
Average	93.3	97.1	86.1	7.3	1.7	13.0	0.0	16.7	23.3	0.96	0.91	0.82



Autoencoder-based one-class classification

Conclusion

- Implemented autoencoder model showed 97.1% maximum accuracy for 8 frontal electrodes configuration with 0.91 AUC value.
- This system can be trained using fewer training trails.
- Weights of trained model can be reused to speed up the training process for new user (transfer learning).



Affect-guided soundscape exploration





Introduction

- Use of sound as a method for reducing stress is common.
- Effectiveness of the sounds used for relaxation depends on personal taste.
- Machine learning search is used to find relaxation “sweet spot” in pantophonic (annular) soundscape.
- Proposed approach has 3 main components
 - Sound generator
 - RL agent
 - EEG analyzer





Methodology: Sound generator

- Max/MSP patch was developed as a soundscape interface.
- Six pantophonic sources are arranged in a ring around the perimeter of a horizontal virtual auditory space.
- Output sound is generated depending on the Cartesian position of the avatar.
- Sound sources are parameterized not only by Cartesian position but also volume, pitch, reverb, and filters.



Methodology: Max/MSP patch (Sound generator)

WASD / JOYSTICK MODE
(point and click disabled)

The diagram illustrates the methodology for the Max/MSP patch. It features a central yellow circle representing the 'Avatar'. Surrounding the avatar are six blue dots, each labeled with a number from 0 to 5. A white box labeled 'Sound sources' has two arrows pointing to dots 0 and 5. Another white box labeled 'Avatar' has an arrow pointing to the central yellow circle. The dots are arranged in a circular pattern around the avatar.

File Selection (drag and drop, or select from a drop menu)

Randomize Pitches Reset All Pitches

0 02 beneath the ice.mp3	1 09 triangle - simple.mp3
2 08 triangle - complex.mp3	3 04 flux.mp3
4 05 frozen tanpura.mp3	5 01 air and steel.mp3
6 07 sharp fifths.mp3	7 03 field crickets.mp3

Each audio source entry includes a waveform visualization and a 'Pitch' control slider.

Playback ON/OFF
(or press SPACEBAR)

Press ⌘- or ⌘+ to resize interface

Shuffle audio sources

Save New Preset

Recall

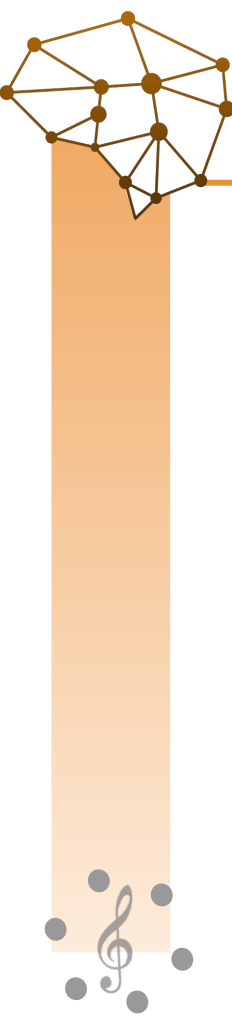
Load Preset

Drag Audio Folder

Files will automatically be assigned to random points in the circle

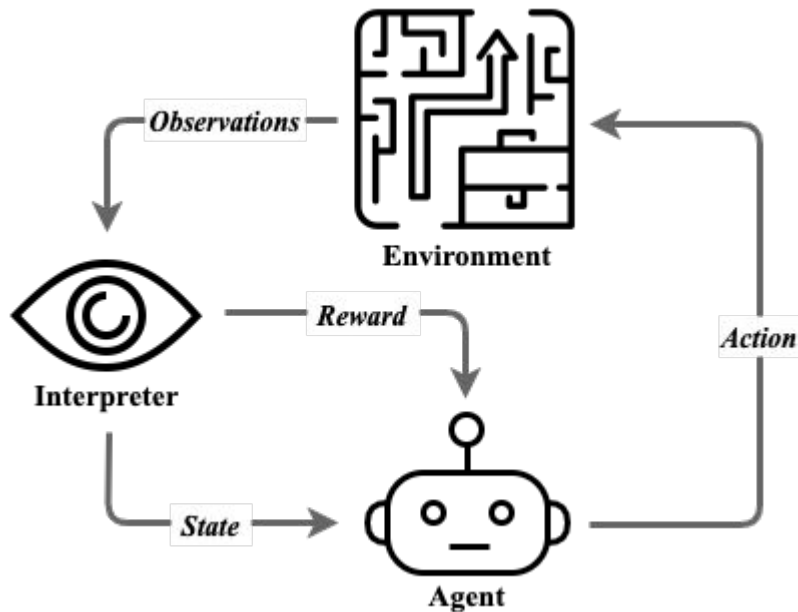
Volume
-6 dB

beta Version 2.1 - July 29 2018



Introduction: RL agent

- Reinforcement learning (RL) is an online learning method that takes action in an environment to maximize cumulative reward.





Methodology: RL agent

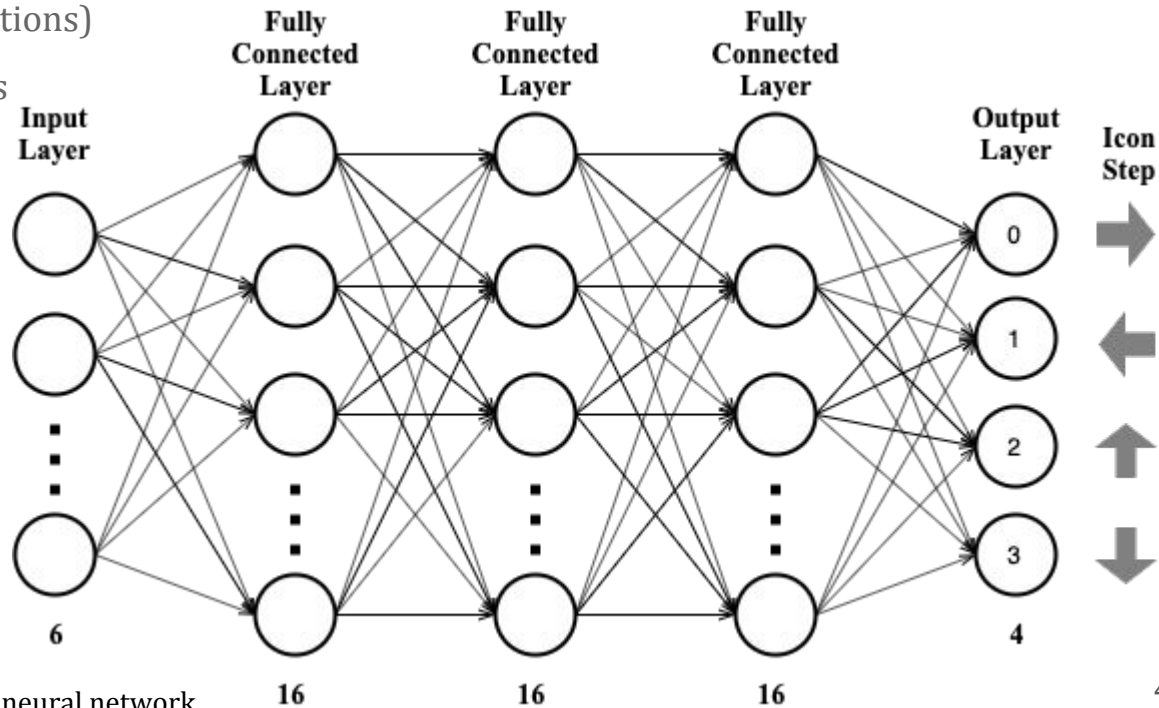
- Keras-rl Python-based library, which uses Tensorflow and Keras, was used to implement RL-agent.
- Keras-rl usually compatible with OpenAI Gym library, which provides video game simulation environments for RL.
- Deep Q Network (DQN) algorithm in RL was used.
- Boltzmann Q policy was used as the policy.
- Adaptive momentum (AdaM) was used as the optimizer.



Methodology: RL agent

- Neural network with 5 fully-connected layers was used

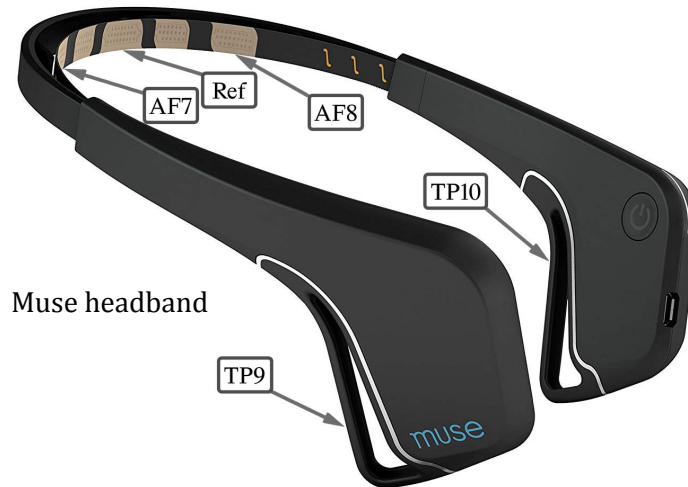
- 6 inputs (observations)
- 4 discrete outputs
- 3 hidden layers



Structure of neural network

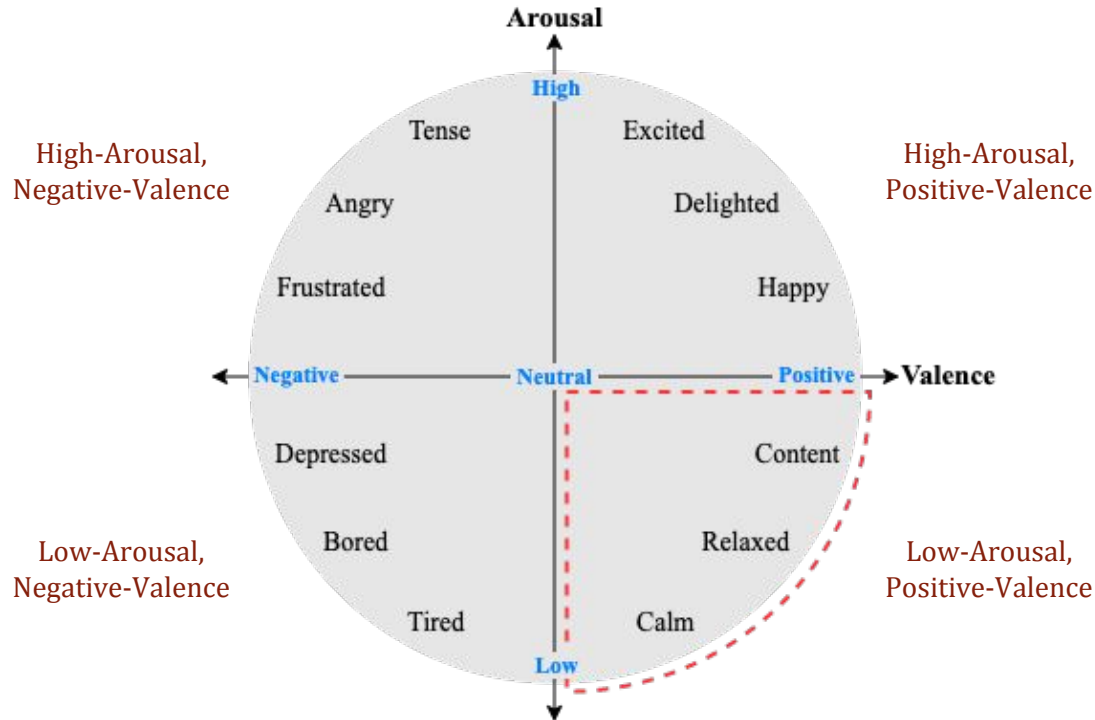
Methodology: EEG analyzer

- EEG signals are used to find the relaxed state of the person (feed as a reward).
- Muse headband, which uses 4 dry electrodes, was used.
- 256 Hz max sampling rate over Bluetooth (BLE)



Methodology: EEG analyzer

- Relaxation state can be recognized in low-arousal, positive-valence region





Methodology: EEG analyzer

- According to Ramirez et al., relaxation state can be recognized using valence-arousal model in passive music therapy.

$$\text{Arousal} = \ln \left(\frac{\beta_{AF7} + \beta_{AF8}}{\alpha_{AF7} + \alpha_{AF8}} \right)$$

$$\text{Valence} = \alpha_{AF8} - \alpha_{AF7}$$

$$\text{Reward} = \ln \left(\frac{\alpha_{AF8}}{\beta_{AF8}} \right) - \ln \left(\frac{\alpha_{AF7}}{\beta_{AF7}} \right)$$

$$= \ln \left(\frac{\alpha_{AF8} \times \beta_{AF7}}{\beta_{AF8} \times \alpha_{AF7}} \right)$$

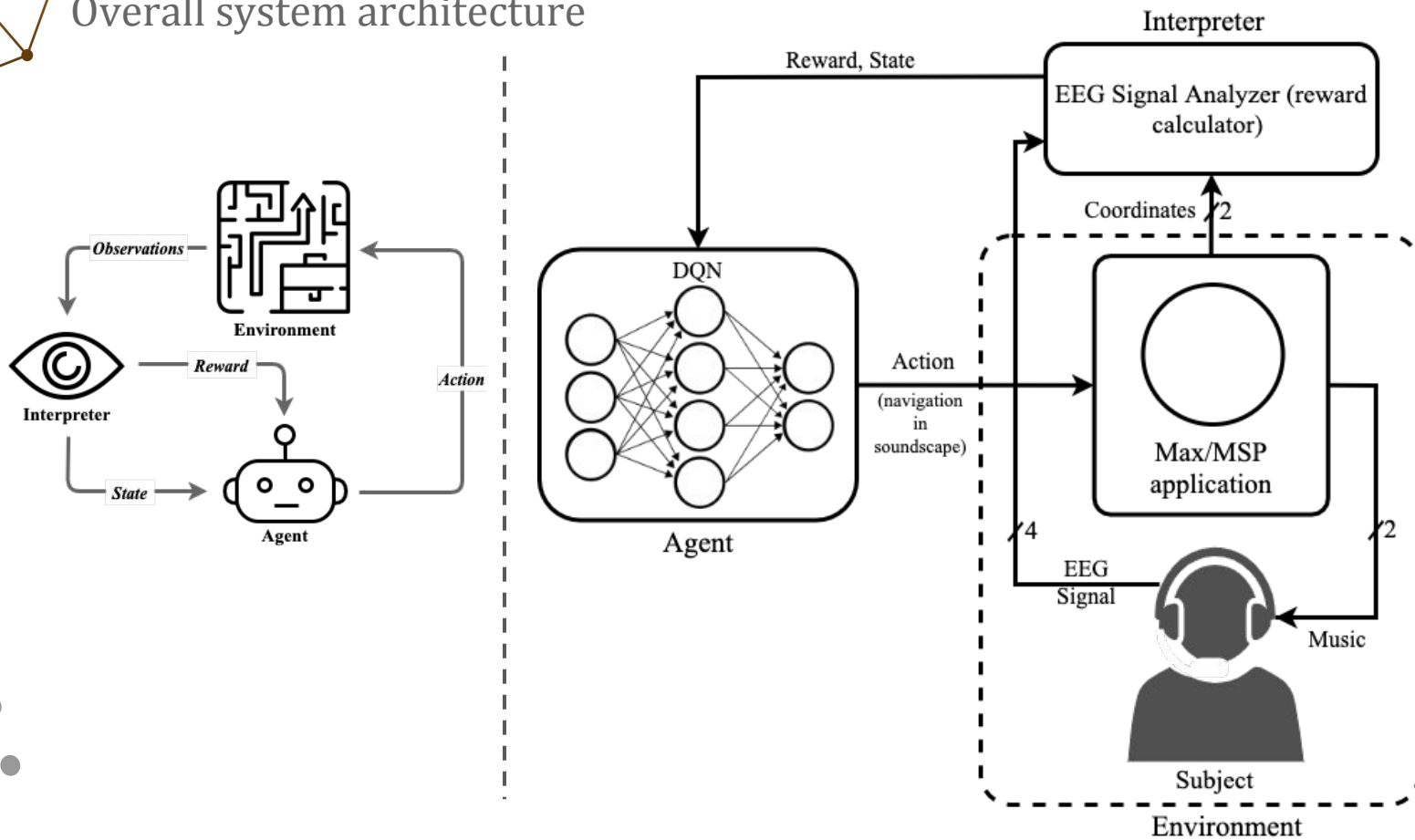



Methodology: EEG analyzer

- Calculated reward value using valence and arousal was fed to RL agent.
 - Cartesian coordinates of avatar (x, y), relative powers of delta (δ), theta (θ), alpha (α), and beta (β) bands were fed to RL-agent as observations.
 - RL-agent output discrete values as cursor movement that can be converted to virtual keystrokes.
 - Position of the avatar was changed by the virtual keystrokes.
 - Core class OpenAI Gym library was extended to have the functionalities of video game environments.
- 

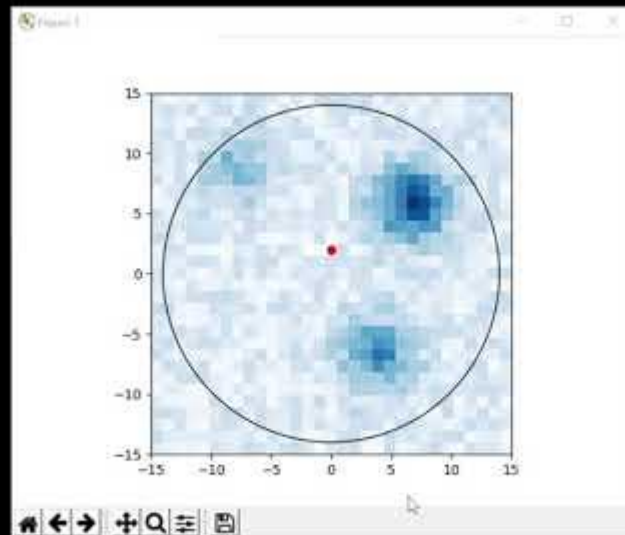
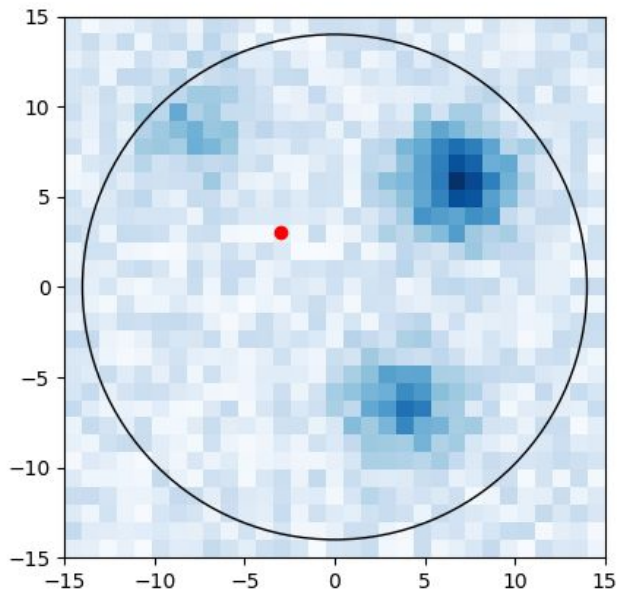
Methodology

Overall system architecture



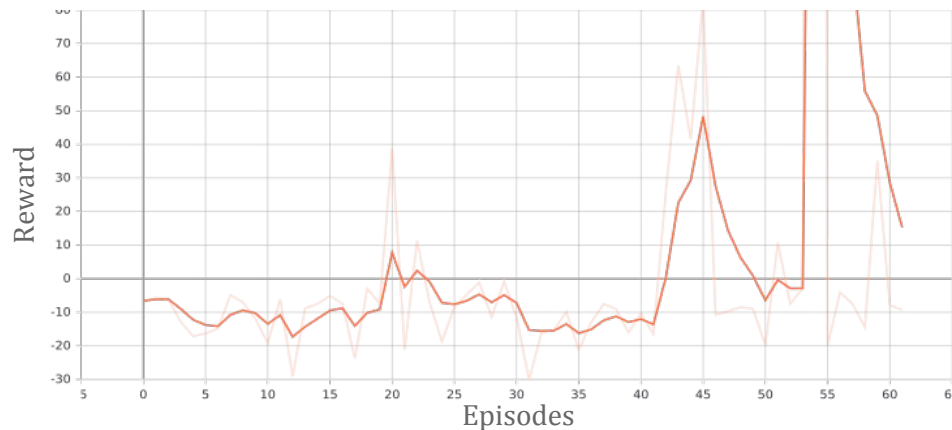
Simulation

- A simulation that can feed pre-defined reward and observations was performed to test performance of the DQN model.

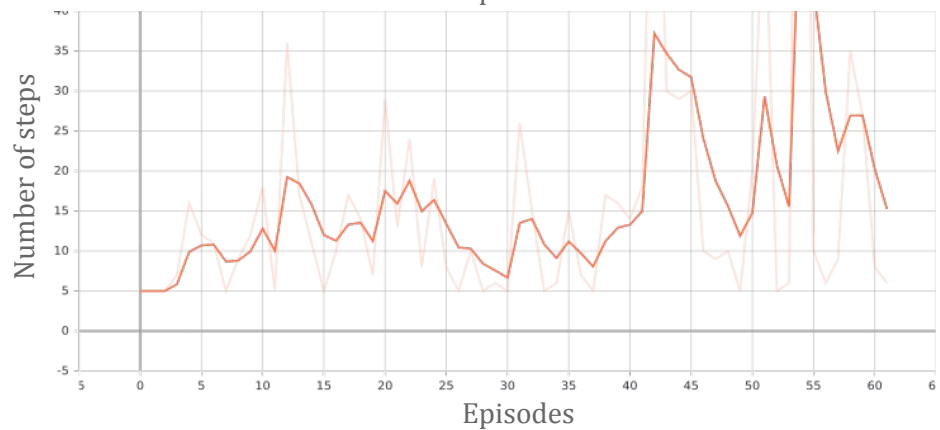


Results of simulation

Results generated by Tensorboard



Variation of reward against episodes

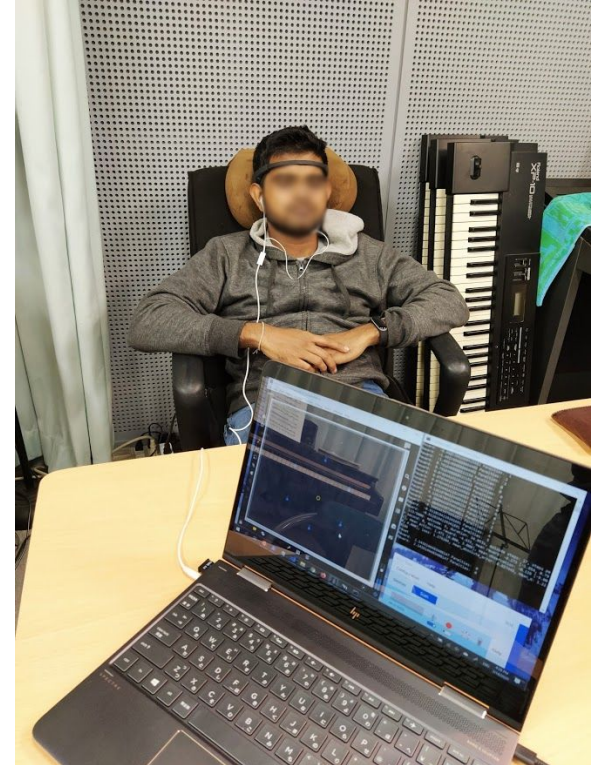


Number of steps against episodes



Experiment with human subjects

- During trials, subjects were instructed to keep eyes closed.
- The realtime data stream was segmented into windows with 50% overlap.
- Butterworth filter used to extract delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), and beta (13–30 Hz) bands.

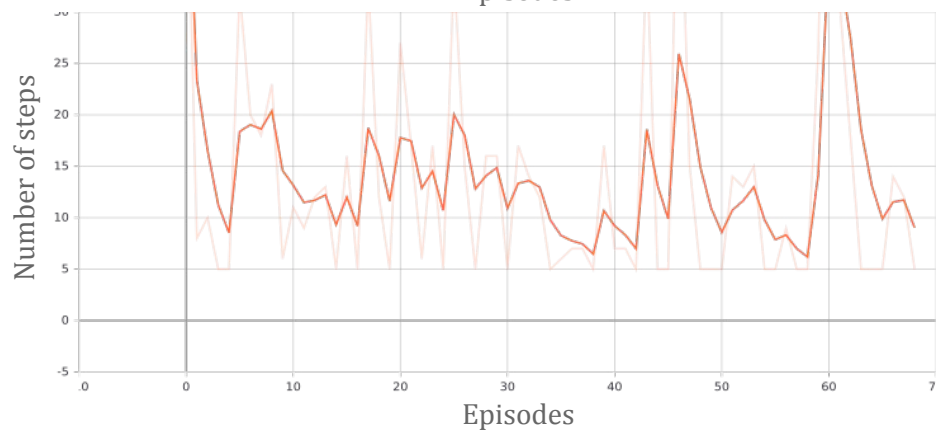


Results of a subject

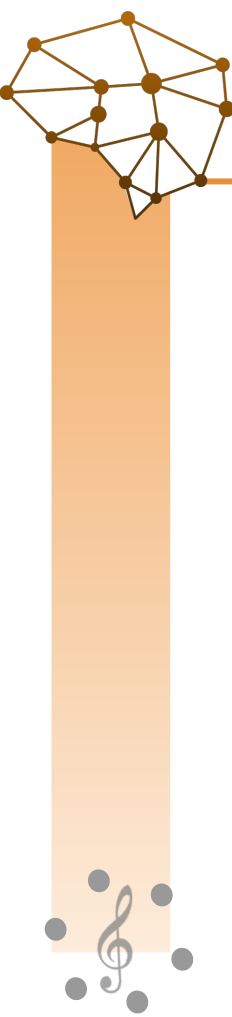
Results generated by Tensorboard



Variation of reward against episodes



Number of steps against episodes



Conclusion

- Relaxation can be described by a stable field across a pantophonic soundscape.
- The sweet spot is probably not a point, but an area or collection of spots, maybe non-contiguous.
- It is unknown how long after the avatar moves and the human hears the new position the brain settles into steady-state response.



Overall Conclusion

- Proposed novel approaches achieved adequate accuracy for user authentication and identification.
- Consumer-grade EEG headsets can be used for authentication.
- Permanence of EEG biometric is still questionable.
- It is possible to use reinforcement learning and EEG data to enhance musical relaxation.





Publications

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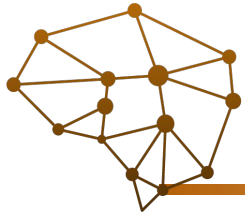
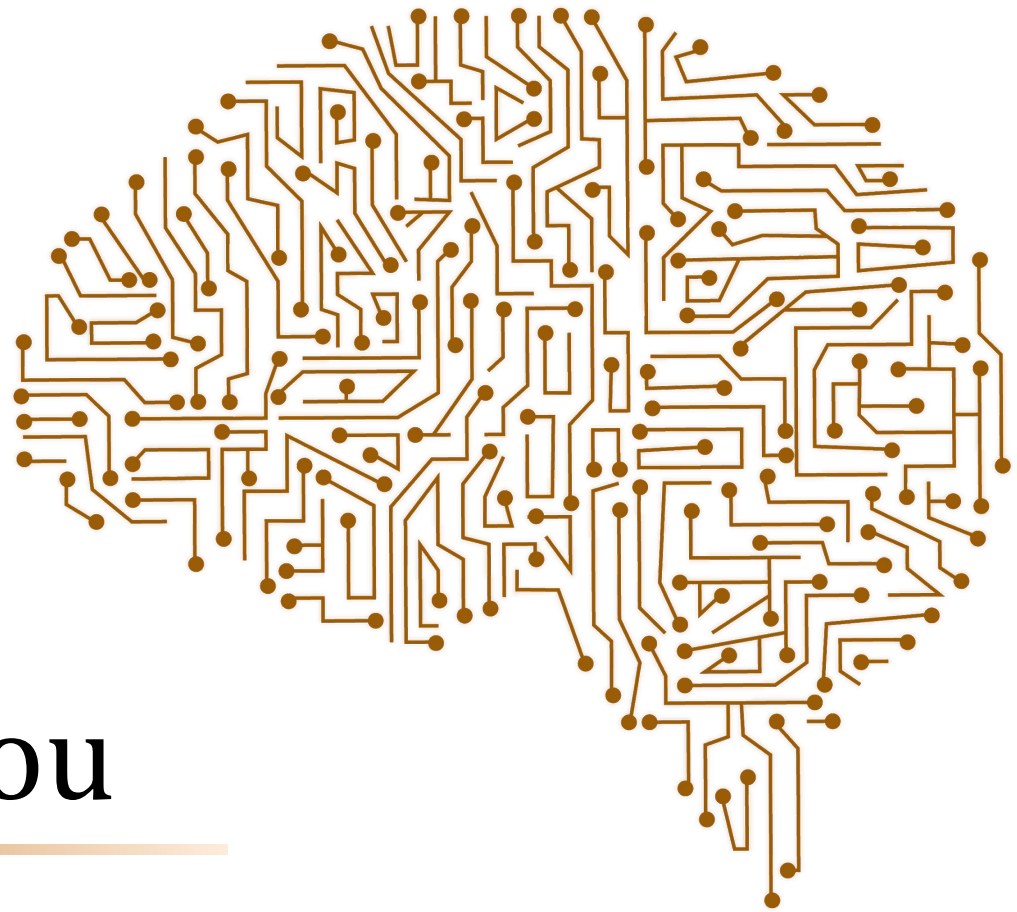
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Thank you
